

1 Solution — GIAM 1.4

Problem 4

1.1 Problem

One method of converting from decimal to another base is *repeated division*: divide the number by the base and record the remainder; then divide the quotient by the base and record the remainder; continue until the quotient is smaller than the base. Convert **3267** to base-**7** using repeated division, and check the answer using the meaning of base-**7** place notation.

1.2 Repeated Division

Divide by **7** over and over, recording each remainder:

Division	Quotient	Remainder
$3267 \div 7$	466	5
$466 \div 7$	66	4
$66 \div 7$	9	3
$9 \div 7$	1	2
$1 \div 7$	0	1

Table 1.1

The remainders, **read from the bottom up** (last computed first), spell out the base-**7** digits:

$$3267 = 12345_7.$$

1.3 The Division Ladder

Repeated division of 3267 by 7:

$$3267 \div 7 = 466$$

$$466 \div 7 = 66$$

$$66 \div 7 = 9$$

$$9 \div 7 = 1$$

$$1 \div 7 = 0$$

(stop when the quotient drops below 7 — the final quotient 1 becomes the leading digit)

remainder

5
4
3
2
1

read the remainders
from the bottom up:

1 2 3 4 5

$$\mathbf{3267 = 12345_7}$$

Check by place notation:

$$\begin{aligned} 12345_7 &= 1 \cdot 7^4 + 2 \cdot 7^3 + 3 \cdot 7^2 + 4 \cdot 7^1 + 5 \cdot 7^0 \\ &= 2401 + 686 + 147 + 28 + 5 = \mathbf{3267} \checkmark \end{aligned}$$

Figure 1.1: Each row divides the previous quotient by 7 and records the remainder (5, 4, 3, 2, 1). Reading those remainders bottom-to-top gives the base-7 digits 1 2 3 4 5, so $3267 = 12345_7$. The check expands $12345_7 = 1 \cdot 7^4 + 2 \cdot 7^3 + 3 \cdot 7^2 + 4 \cdot 7 + 5 = 3267$.

1.4 Checking with Place Notation

Expand 12345_7 by the meaning of base-7 place notation and confirm it returns **3267**:

$$\begin{aligned} 12345_7 &= 1 \cdot 7^4 + 2 \cdot 7^3 + 3 \cdot 7^2 + 4 \cdot 7^1 + 5 \cdot 7^0 \\ &= 1 \cdot 2401 + 2 \cdot 343 + 3 \cdot 49 + 4 \cdot 7 + 5 \cdot 1 \\ &= 2401 + 686 + 147 + 28 + 5 \\ &= \mathbf{3267.} \quad \checkmark \end{aligned}$$

The conversion checks out: $\mathbf{3267 = 12345_7}$.

1.5 Remark — Why the Remainders Are Read Backwards

The first division produces the *units* digit, but we want to *write* the number most-significant digit first — so the order of writing is opposite to the order of computing. Here is why the first remainder is the units digit. Writing the place-value expansion and factoring out a 7:

$$3267 = \underbrace{(1 \cdot 7^3 + 2 \cdot 7^2 + 3 \cdot 7 + 4)}_{\text{quotient 466}} \cdot 7 + \underbrace{5}_{\text{remainder}} .$$

The remainder of $3267 \div 7$ is exactly the units digit (5), and the quotient 466 “holds” all the higher digits, *shifted down by one place*. Dividing again peels off the next digit (4), and so on. The last division exposes the leading digit. So the remainders emerge units-first; reading them bottom-up restores the proper left-to-right order.

1.6 Remark — Why We Stop When the Quotient Drops Below the Base

The instruction says to stop once the quotient is smaller than the base. At the step $9 \div 7 = 1$ remainder 2, the quotient 1 is already less than 7, so 1 cannot be split further — it *is* a single base-7 digit, the leading one. (Equivalently, one more division $1 \div 7 = 0$ remainder 1 makes the quotient 0; the table above shows this final step explicitly, with the remainder 1 as the leading digit.) Either way the process terminates because the quotients strictly *decrease* at every step — they shrink by a factor of about 7 each time — and a strictly decreasing sequence of non-negative integers must reach 0.

1.7 Remark — The Memorable Answer 12345_7

It is a pleasant coincidence that 3267 converts to the consecutive digits 12345_7 . This is the base-7 analogue of asking “what decimal number is 12345?” — the place-value expansion $1 \cdot 7^4 + 2 \cdot 7^3 + 3 \cdot 7^2 + 4 \cdot 7 + 5$ just happens to equal 3267. (In decimal, $12345 = 1 \cdot 10^4 + \dots + 5$ is of course 12,345 — much larger, because base $10 > 7$ weights each place more heavily.) The neat answer is a good final sanity check: a string of *increasing* digits like 1, 2, 3, 4, 5 is easy to verify both ways.

1.8 Remark — The Method Works for Any Base

Nothing about repeated division is special to 7. To convert a decimal integer N to base b , repeatedly divide by b and read the remainders bottom-up; each remainder lies in $\{0, 1, \dots, b-1\}$, exactly the digit-set of base b . For example, the same 3267:

- in base 2: $3267 = 110011000011_2$;
- in base 8: $3267 = 6303_8$;
- in base 16: $3267 = \text{CC}3_{16}$ (using $\text{C} = 12$).

The repeated-division algorithm and its place-notation inverse are a matched pair: division *extracts* digits one at a time, place notation *reassembles* them. Together they let us move a number freely between any two bases (routing through decimal, as the textbook suggests, or directly).

1.9 Remark — How Many Digits to Expect

The number of base- b digits of N is $\lfloor \log_b N \rfloor + 1$ — equivalently, the number of repeated-division steps before the quotient hits 0. For $N = 3267$, $b = 7$:

$$\log_7 3267 = \frac{\ln 3267}{\ln 7} \approx \frac{8.092}{1.946} \approx 4.16,$$

so $\lfloor 4.16 \rfloor + 1 = 5$ digits — matching the five digits of 12345_7 and the five rows of the division table. A smaller base needs *more* digits (base 2 took twelve), a larger base *fewer* (base 16 took three): the digit count scales like $1/\log b$, the same reason hexadecimal is a compact way to write long binary strings.